1. Worst-case complexity derivation

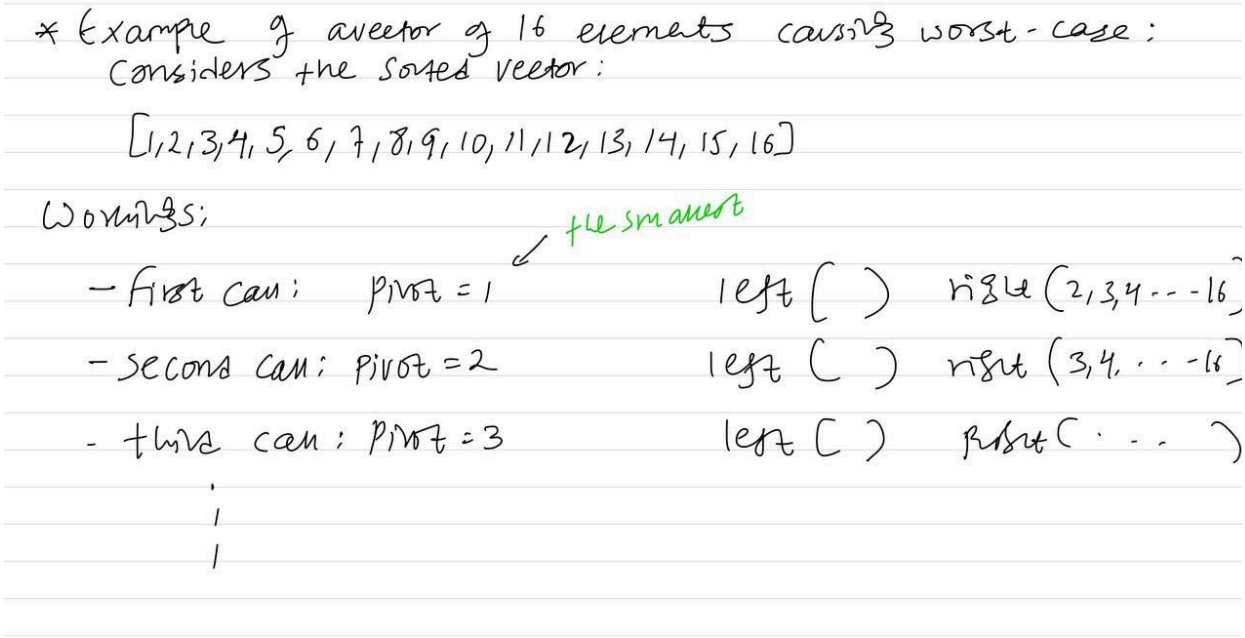
In the worst-case, the pivot is always the smallest element, so one partition is empty and the other contains (n-1) elements. This gives the recurrence

$$T(n) = T(n-1) + O(n)$$

Straighten the recurrence results in:

$$T(n) = O(n) + O(n-1) + \ldots + O(1) = O(n^2)$$

2.



3. Measurement results

We ran quicksort in worst-case inputs (i.e., sorted arrays) for increasing sizes. The measured execution times increase roughly quadratically with n.

4. Plot discussion

We ran quicksort in worst-case inputs (i.e., sorted arrays) for increasing sizes. The measured execution times increase roughly quadratically with n. The plot shows a clear quadratic trend in execution time, matching the O(n^2)worst-case complexity derived earlier.